Execution records, deviations, and ladder maintenance

> **Reading note.** This is an explanatory reconstruction of Jiaxuan Zou's literature synthesis. Published experiments belong to the cited teams. Numbered protocols and illustrative matrices are our planning synthesis unless explicitly attributed.

The blog's §12 turns experiments into reusable evidence. A fit cannot be audited if the run table loses a dataset version, numerical change, failed job, or tuning decision.

Minimum record

Artifact	Required contents
Run manifest	Commit, data/tokenizer versions, architecture, optimizer, schedule, seeds
Measurements	Parameter/FLOP definitions, token counts, validation versions
Runtime record	Hardware, precision, parallelism, throughput, failures, restarts
Lineage	Parent checkpoint, shared prefix, annealing branch, stage transitions
Fit record	Input run IDs, objective, exclusions, solver starts, uncertainty procedure
Decision record	Predicted gain, interval, acceptance rule, untouched holdout result

Procedure

1. Create the manifest before launch and preserve it with the run artifacts.
2. Log deviations as events. Do not silently replace the original configuration after a restart.
3. Keep failed runs in the ledger, distinguishing hardware failures from recipe instability.
4. Reproduce the final fit from a versioned, frozen run table.
5. Preserve the original decision and the evidence available at that time.
6. When data, architecture, optimizer, tokenizer, or precision changes, run a transfer check before reusing an old law.

A source snapshot is also an artifact: record paper identifier/version, retrieval date, PDF, LaTeX availability, and code commit or archive URL. A missing source or code repository should be marked unavailable, not represented by an empty success directory.

The accompanying repository collector records download status in each reference's `metadata.json`. A repository linked by a paper can be a baseline, dataset, or evaluation dependency. Its role must be inspected before calling it the paper's implementation.

Acceptance checklist: complete run lineage; reproducible fit inputs; failures accounted for; holdout usage documented; source versions retained; known implementation mismatches listed.

Sources: blog §12; Marin scaling-heuristic recipe.

Blog/notes crosswalk · Detailed companion